

PROPAGATION DELAY ANALYSIS OF FULL ADDER

1)INPUT A---> SUM

$$T_{\text{delay_sum}} = (t_{\text{phl}}(\text{sum}) + t_{\text{lp}}(\text{sum})) / 2;$$

$$T_{50\%} = 5.0 / 2 = 2.5V;$$

$$T_{\text{phl}}(\text{sum}) = 154.934172\text{p}; (\text{from the graph})$$

$$T_{\text{lp}}(\text{sum}) = 241.93658\text{p}; (\text{from the graph})$$

$$T_{\text{delay_sum}} = (154.934172\text{p} + 241.93658\text{p}) / 2 = \mathbf{198.43534\text{p}};$$

2)INPUT A---> CARRY

$$T_{50\%} = 2.5\text{v};$$

$$T_{\text{phl}}(\text{carry}) = 272.674667\text{p}; (\text{from the graph});$$

$$T_{\text{lp}}(\text{carry}) = 43.471591\text{p}; (\text{from the graph});$$

$$T_{\text{delay_carry}} = (272.674667\text{p} + 43.471591\text{p}) = \mathbf{158.0731285\text{p}}$$